Predictive Modelling of Glioma Using Machine Learning

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***Abstract*— Gliomas, being highly aggressive brain tumours, necessitate accurate grading to guide treatment strategies and predict patient outcomes. Although current grading methods rely on subjective and invasive histopathological examination and fail to fully capture tumour heterogeneity, machine learning (ML) offers a promising alternative. This study explores the application of various ML models, integrating clinical and molecular features for non-invasive glioma grading. Employing models such as Neural Networks, Support Vector Machines, and Stacking, we achieved high accuracy, with notable improvements post-SMOTE implementation to address class imbalance. For instance, the Neural Network model demonstrated the highest accuracy (0.9128), precision (0.9194), and recall (0.9107), significantly surpassing other models like KNN and Decision Trees in performance metrics. Comparative analysis with previous studies reveals that our SVM model, with and without feature selection (FS), outperforms earlier implementations, showing an accuracy of 90.26% compared to 86.769% in prior SVM applications without FS. This indicates robust improvements in predictive capabilities through refined feature selection, highlighting the critical roles of IDH1 and PTEN genes. Despite these advancements, challenges such as data quality and model interpretability persist, necessitating future research focused on model validation in diverse patient populations and enhancement of interpretability for clinical acceptance. Our findings suggest that ML can significantly enhance the precision of glioma grading, thereby potentially improving treatment outcomes in neuro-oncology.**

Keywords*— *Neural Network, Support Vector Machine, Ensemble Models Stacking, Accuracy, Recall

I. INTRODUCTION

Machine learning is significantly changing how we diagnose diseases, especially in the field of cancer, such as glioma, which is a commonly occurring and severe type of brain tumour [1]. Typically, doctors classify gliomas by examining them under a microscope. This examination is crucial because it helps determine the most effective treatment plan for patients. However, this traditional method has several drawbacks. It can be invasive, requiring tissue samples, and it often relies heavily on individual judgment, which can vary between different observers [2]. Machine learning offers a promising alternative by potentially providing more accurate and consistent diagnostic results.

By training machine learning algorithms with a wide array of data, including clinical information, molecular, and genetic details from patients, we can detect patterns that might not be visible through traditional methods. This approach promises a less invasive and more objective way to grade gliomas. Ultimately, this could lead to a more refined and accurate grading system, enabling doctors to make more informed decisions regarding treatment options.

A. The Problem

Gliomas are among the most common and aggressive brain tumours, classified into different grades [3] based on their histological characteristics. The grading of gliomas is crucial as it directly influences the treatment strategy and the prognosis for patients. Currently, the grading is primarily done through histopathological examination, an approach that, while effective, has its limitations [4]. The subjective nature of this method can lead to inter observer variability, and being invasive, it carries inherent risks and limitations.

Moreover, the histopathological approach may not fully capture the tumour's heterogeneity, potentially leading to under or overestimation of the tumour grade. Accurate grading is essential, as low-grade gliomas (LGG) and high-grade gliomas, such as Glioblastoma Multiforme (GBM), require distinctly different treatment approaches and have vastly different prognostic outcomes. Early and precise grading allows for more targeted therapies, which can significantly improve survival rates and quality of life in patients. As reported by L. Capelle et al., targeted therapies based on precise grading have demonstrated improved survival outcomes in patients with LGG [5]. The need for a more objective, reliable, and non-invasive grading method is evident, with machine learning offering a promising solution. By leveraging complex datasets that include clinical and molecular information, machine learning can provide a nuanced and comprehensive analysis, potentially surpassing traditional methods in accuracy and reliability. B.J. Erickson et al. have highlighted that machine learning models, particularly those employing deep learning, offer enhanced accuracy and reliability over traditional histopathological methods in grading gliomas [6].

B. The Identified Dataset

The dataset identified for this project The dataset used "Glioma Grading Clinical and Mutation Features," available from the UCI Machine Learning Repository [7], is an extensive compilation of data collected from 839 glioma patients. It uniquely combines clinical and molecular information, making it a valuable resource for research. The dataset encompasses a wide array of variables that are meticulously grouped into independent and dependent categories, providing a robust foundation for analytical models.

Independent Variables: The dataset includes basic demographic details such as age and gender of the patients. This helps in understanding the distribution of glioma across different age groups and between genders. Molecular data

includes comprehensive genetic information like mutations in 20 significant genes known to play a role in glioma pathogenesis and their expression levels. Such molecular data is pivotal for understanding the genetic drivers of the disease and exploring targeted therapies.

Dependent Variable: The primary outcome variable is the glioma grade, which classifies each tumour into categories such as low-grade glioma (LGG) or Glioblastoma Multiforme (GBM). This classification is based on histological examination according to established World Health Organization (WHO) criteria. Determining the glioma grade is essential for predicting patient outcomes and guiding treatment decisions.

The dataset is derived from the broader studies conducted by The Cancer Genome Atlas (TCGA), specifically the TCGA-LGG and TCGA-GBM projects [8]. These projects are well-known for their thorough data collection and stringent validation processes, enhancing the dataset's credibility. The integration of such high-quality data allows for the application of advanced machine learning techniques not only to predict glioma grades with greater accuracy but also to uncover new biomarkers and genetic profiles linked to prognosis and treatment responsiveness.

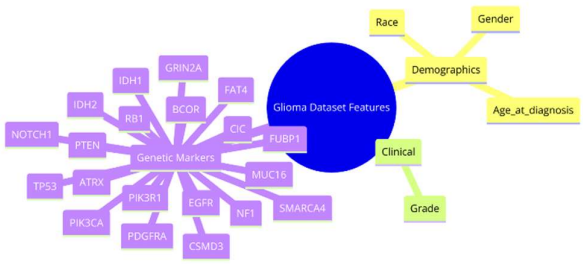


Fig 1 Feature Map of Glioma Dataset

C. Existing Work

The application of machine learning in medical imaging and diagnosis has grown exponentially, with glioma grading being a significant area of focus. Research in this field has leveraged various machine learning algorithms to enhance the accuracy and efficiency of glioma classification. Studies have utilized a range of techniques from traditional statistical models like logistic regression to more complex models like deep neural networks. These studies have shown that machine learning can achieve high levels of accuracy, often surpassing traditional methods in both speed and reliability.

A recent study employed a convolutional neural network (CNN) to classify MRI images of gliomas into different WHO grades with an accuracy exceeding 90%. [7] This model was trained on a large dataset of MRI scans, where it learned to recognize patterns and features associated with different tumour grades. The success of such models highlights the potential of machine learning to support radiologists and pathologists by providing a second, highly reliable opinion that can help reduce subjective discrepancies in tumour grading.

Moreover, beyond image-based analyses, machine learning techniques have been applied to genetic and molecular data to predict glioma grades. These models analyse patterns in gene expression, mutations, and other molecular markers to classify tumours. For instance, random forest classifiers and support vector machines have been used to differentiate between LGG and GBM based on genetic profiles, achieving accuracies ranging from 70% to 86% [8]. These results underscore the capability of machine learning to integrate complex biological data sets and derive meaningful predictions that can guide clinical decision making.

Despite these advances, challenges remain in the field. Many models suffer from overfitting, where they perform well on training data but fail to generalize to new, unseen data. Additionally, the interpretability of machine learning models, particularly deep learning models, is often limited making it difficult for clinicians to understand the basis of the model's predictions. Addressing these challenges is crucial for the wider acceptance and clinical adoption of machine learning in glioma grading.

II. METHODOLOGY

In this project, Python was extensively used for data analysis and model development. Key libraries such as Pandas and NumPy facilitated data handling, while Matplotlib and Seaborn enabled effective data visualization Machine learning models, including Random Forest and SVM, were built and evaluated using scikit-learn, with XGBoost enhancing model performance. Additionally, SMOTE was employed to address data imbalance. This comprehensive toolkit ensured efficient data processing, robust model development, and thorough evaluation, effectively meeting the research objectives.

A. Business Understanding

The primary objective of this project is to develop a predictive model capable of accurately grading gliomas using clinical and molecular features, thereby aiding clinicians in making informed treatment decisions. This objective arises from the critical need to address the limitations of current glioma grading methods, mainly their subjectivity and invasiveness. By applying machine learning techniques, this project aims to provide a more objective and non-invasive alternative for grading gliomas, which could lead to more consistent and personalized treatment strategies.

The expected outcome of this project is a machine learning model that not only predicts glioma grades with high accuracy but also offers insights into the most predictive features, thus enhancing our understanding of glioma biology. Such a model could be integrated into clinical settings, providing oncologists with an additional diagnostic tool that complements traditional histopathological techniques. This integration could potentially increase the accuracy of glioma grading, facilitate earlier and more precise treatment interventions, and ultimately improve patient outcomes.

B. Data Understanding

The dataset, "Glioma Grading Clinical and Mutation Features" is a multi-dimensional dataset providing a holistic view of patients diagnosed with glioma. It encapsulates various attributes from basic demographic details to complex molecular and genetic markers.

The dataset comprises clinical features like:

- Grade: Categorization of tumours into Low-Grade Glioma (LGG) or Glioblastoma (GBM), serving as the outcome variable for predictive modelling efforts.
- Gender: Gender of the patient.
- Age_at_diagnosis: Age of the patient at the time of diagnosis.
- Race: Ethnicity or race of the patient.
- IDH1: Mutation status of the IDH1 gene, a crucial marker in glioma prognosis.
- TP53: Mutation status of the TP53 gene, a key tumour suppressor.
- ATRX: Mutation status of the ATRX gene, associated with alternative lengthening of telomeres in certain gliomas.
- PTEN: Mutation status of the PTEN gene, involved in cell growth regulation.
- EGFR: Mutation status of the EGFR gene, associated with cell signalling and growth in gliomas.
- CIC: Mutation status of the CIC gene, involved in cell cycle regulation.
- MUC16: Mutation status of the MUC16 gene, encoding a protein involved in cell adhesion.
- PIK3CA: Mutation status of the PIK3CA gene, involved in the PI3K/AKT signalling pathway.
- NF1: Mutation status of the NF1 gene, encoding a tumour suppressor protein.
- PIK3R1: Mutation status of the PIK3R1 gene, encoding a regulatory subunit of PI3K.
- FUBP1: Mutation status of the FUBP1 gene, encoding a DNA-binding protein.
- RB1: Mutation status of the RB1 gene, encoding a tumour suppressor protein.
- NOTCH1: Mutation status of the NOTCH1 gene, involved in cell differentiation and survival.
- BCOR: Mutation status of the BCOR gene, encoding a transcriptional co-repressor.
- CSMD3: Mutation status of the CSMD3 gene, encoding a protein involved in complement regulation.
- SMARCA4: Mutation status of the SMARCA4 gene, encoding a subunit of the SWI/SNF chromatin remodelling complex.
- GRIN2A: Mutation status of the GRIN2A gene, encoding a subunit of the NMDA receptor.
- IDH2: Mutation status of the IDH2 gene, encoding an isocitrate dehydrogenase enzyme.
- FAT4: Mutation status of the FAT4 gene, encoding a cadherin-related protein.
- PDGFRA: Mutation status of the PDGFRA gene, encoding a platelet-derived growth factor receptor.

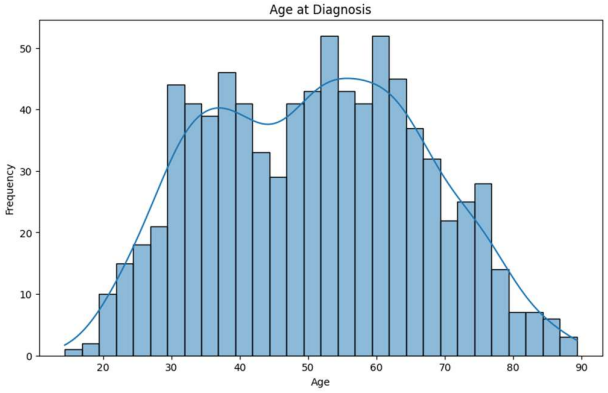


Fig 2 Histogram of Age at Diagnosis

This histogram illustrates the distribution of ages among patients diagnosed with glioma, providing insights into the age demographics of the study population.

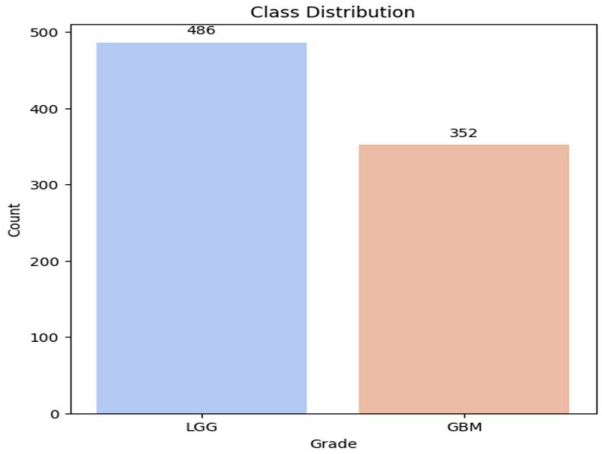


Fig 3 Grade Class Distribution

This visualization showcases the distribution of glioma grades (LGG vs. GBM) within the dataset, highlighting the imbalance between different classes.

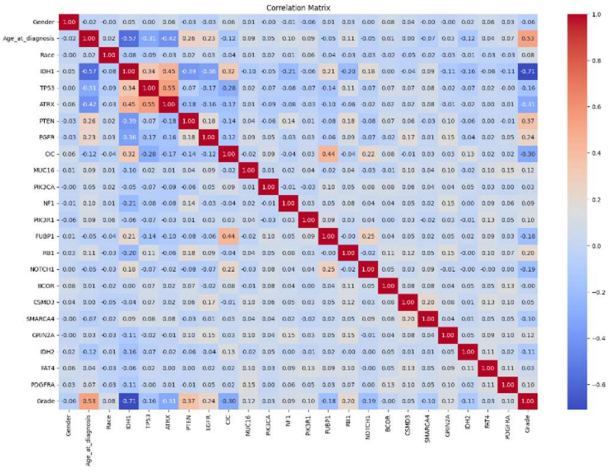


Fig 4 Correlation Matrix

The correlation matrix depicts the relationships between various features in the dataset, shedding light on potential associations and dependencies among variables.

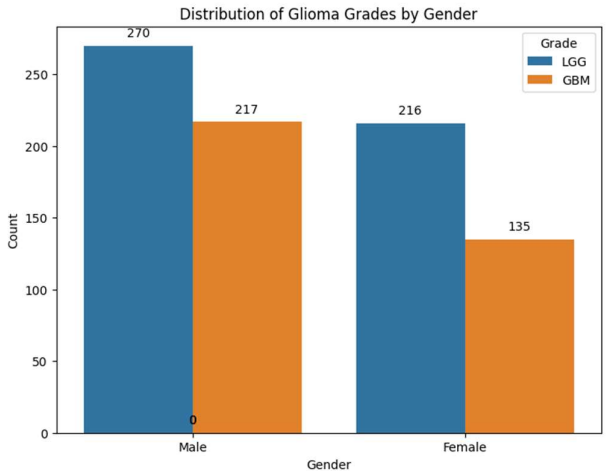


Fig 5 Distribution of Glioma Grades by Gender

This visualization explores the distribution of glioma grades across different genders, offering insights into potential gender-based disparities in tumour grading.

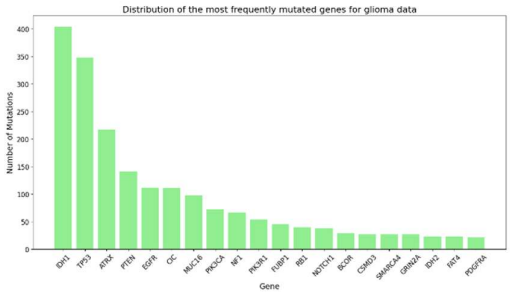


Fig 6 Distribution of the most frequently mutated genes

The graph in figure 6 show the distribution of mutations across the top genes, helping to identify which genes are most affected in glioma cases according to the TCGA data.

C. Data Preparation

The data preparation process involves cleaning and organizing the glioma dataset to ensure it's suitable for machine learning modelling. Initially the dataset undergoes a cleaning phase where any rows with missing values are eliminated, ensuring the data's integrity and consistency for analysis. In this specific case, the dataset contains no missing values, indicating that all records are complete with all necessary information. Furthermore, identify and removes duplicate entries, maintaining the dataset's uniqueness and reliability. It is noted that there was one duplicate record found and removed, which is crucial for preventing the model from being biased or overfitting due to repeated data points. The dataset is then split into features and the target variable, 'Grade', which represents the glioma grades. The Synthetic Minority Over Sampling Technique (SMOTE) is applied to address any class imbalance by generating synthetic samples for the minority class, ensuring the model is not biased towards the majority class. In this instance, the class imbalance before applying SMOTE was significant, with a 48:52% distribution, which could have led to a model biased towards the majority class. Finally, the data is divided into training and testing sets, enabling the evaluation of the model's performance on unseen data, a crucial step to assess its predictive capabilities and generalization.

D. Modelling

In selecting the machine learning algorithms for glioma prediction, several factors were considered, including the nature of the dataset, the complexity of the problem, and the specific requirements of the predictive modelling task.

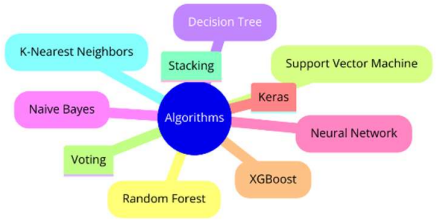


Fig 7 Modelling Algorithms

i KNN (K-NEAREST NEIGHBORS):

KNN was chosen for its simplicity and ability to identify patterns based on the closeness of data points. It's particularly useful for categorizing tumours based on the similarity of clinical and genetic features to other known cases. While there may not be specific historical studies focusing solely on glioma prediction with KNN, its effectiveness in pattern recognition tasks has been demonstrated in various medical domains [9].

ii Random Forest:

Random Forest was selected for its robustness and ability to handle complex datasets. It's capable of analysing the importance of different features and identifying influential factors in tumour behaviour. Previous studies have shown the effectiveness of Random Forest in medical diagnosis and prognosis prediction tasks, including glioma grading [10].

iii SVM (Support Vector Machine):

SVM was chosen for its effectiveness in high-dimensional spaces and ability to separate different classes. It's well-suited for datasets with many features, such as those including detailed genetic information. Studies have demonstrated the utility of SVM in various medical applications, including tumour classification and prognosis prediction [11].

iv Decision Tree:

Decision Tree was selected for its ability to provide a clear visualization of decision paths. It's useful for understanding the relationships between different features and tumour classifications. While not specific to glioma grading, Decision Trees have been widely used in medical decision support systems and have shown promising results in various clinical prediction tasks [12].

v Naive Bayes:

Naive Bayes was chosen for its simplicity and efficiency, particularly in handling categorical data. It's suitable for quick assessments of tumour grade based on observable clinical signs and genetic mutations. Previous studies have demonstrated the effectiveness of Naive Bayes in medical diagnosis and classification tasks [13].

vi Neural Network:

Neural Networks were selected for their ability to model complex, nonlinear relationships. They're powerful for capturing subtle patterns within medical data and predicting tumour grades with high accuracy. Numerous studies have shown the efficacy of Neural Networks in medical image analysis, diagnosis, and prognosis prediction tasks [14].

vii XGBoost:

XGBoost was chosen for its speed, performance, and ability to handle large and complex datasets. It's effective in enhancing predictive accuracy and speed in identifying tumour grades. Studies have demonstrated the superiority of XGBoost in various machine learning competitions and medical applications [15].

viii Ensemble Models:

Ensemble models were selected to leverage the strengths of multiple algorithms and improve overall prediction accuracy. Stacking [16]and Voting [17] ensembles were chosen for their ability to capture complex interdependencies between predictions and blend insights from different models. While not specific to glioma grading, ensemble methods have been widely used in medical decision support systems and have shown promising results in various clinical prediction tasks [18] .

E. Evaluation

Model evaluation will focus on several key metrics to assess the performance of the machine learning models:

Accuracy: While a general measure of model performance, accuracy will be supplemented with more detailed metrics due to its limitations in the context of imbalanced datasets.

Precision, Recall, and F1 Score: These metrics will help evaluate the model's ability to classify the glioma grades correctly without disproportionately misclassifying one grade as another.

ROC Curve: This will provide a comprehensive view of model performance across different classification thresholds, offering insights into the trade-off between true positive rates and false positive rates.

Confusion Matrix: An essential tool for understanding the classification performance for each class, helping to pinpoint weaknesses in the model, such as a tendency to misclassify one type of glioma as another.

Besides these quantitative metrics, the models will be evaluated based on their interpretability and the clinical relevance of their findings. The goal is to select a model that not only performs well statistically but also makes sense from a medical perspective, providing insights that could potentially lead to better clinical decision making.

III. RESULTS

Various machine learning models have been applied to enhance the diagnosis and classification of Glioma grades. These models are used to identify patterns and predict outcomes from the molecular data. This report provides a comprehensive overview of the performances of ten different models: KNN, Random Forest, Support Vector Machine (SVM), Decision Tree, Naive Bayes, Neural Network, Keras, XGBoost, Voting, and Stacking models, each evaluated across multiple metrics: accuracy, precision, recall, F1 score, ROC AUC, and mean absolute error (MAE).

A. Original Data Evaluation

The SVM Model and the Stacking Model are indeed top performers with identical metrics, both achieving 87.5% accuracy, 87.28% precision, an 88.85% recall, and 88.85% ROC AUC scores, as well as sharing the lowest MAE at 12.5%. It is important to emphasize that both the SVM and Stacking Model exhibit strong performance, not just the SVM alone.

The Decision Tree Model did show the lowest scores for most metrics; however, it is noteworthy that the Naive Bayes Model had a slightly lower F1-Score than the Decision Tree Model (0.8330 compared to 0.7966).

The Naive Bayes Model had a high recall of 85.89%, not the highest among the models. Both the SVM and Stacking Models had a higher recall at 88.85%. Precision for the Naive Bayes Model is also not the lowest; the Decision Tree Model has a lower precision.

When discussing the Neural Network and Keras Model, it should be mentioned that their accuracy is around 85.71% and 83.93%, respectively, not 87.5% and 87.8%. While these models perform well, they do not match the top accuracy of the SVM and Stacking Models.

XGBoost has shown solid performance but with slightly lower metrics compared to the SVM and Stacking Models, contrary to the claim that it performed similarly to the Random Forest. It's worth noting that it did have a lower MAE than the RF Model.

The Voting Model shows strong performance with an accuracy of 86.9%, but not as high as the SVM and Stacking Models. The analysis should be adjusted to reflect that it's not just the Stacking Model that has one of the highest performances, but the Voting Model does well too.

The analysis is correct in stating that the highest ROC AUC scores are seen in the SVM and Stacking Models, indicating strong discriminatory power.

The Mean Absolute Error (MAE) was indeed the lowest for the SVM and Stacking Models, which should be highlighted as a measure of their reliability in predictions.

In summary, the SVM and Stacking Models are indeed recommended for deployment, but it should be clearly stated that both models share the top spot in performance based on the metrics provided.

TABLE 1. THE COMPUTATIONAL RESULTS WITHOUT THE FEATURE SELECTION PROCESS

Models	Accuracy	Precision	Recall	F1-Score	ROC AUC	MAE:
KNN Model	0.8214	0.8165	0.8289	0.8183	0.8289	0.1786
RF Model	0.8393	0.8319	0.8412	0.8350	0.8412	0.1607
SVM Model	0.8750	0.8728	0.8885	0.8734	0.8885	0.1250
Decision Tree	0.8036	0.7949	0.7990	0.7966	0.7990	0.1964
Naive Bayes	0.8333	0.8482	0.8589	0.8330	0.8589	0.1667
Neural Network	0.8571	0.8540	0.8686	0.8551	0.8686	0.1429
Keras Model	0.8393	0.8393	0.8538	0.8376	0.8538	0.1607
XGBoost	0.8631	0.8577	0.8711	0.8605	0.8711	0.1369
Voting Model	0.8690	0.8659	0.8810	0.8672	0.8810	0.1310
Stacking Model	0.8750	0.8728	0.8885	0.8734	0.8885	0.1250

B. Synthetic Data Evaluation

SVM Model and Stacking Model: These two models perform equally well, with the second highest accuracy (90.26%), precision (91.13%), recall (90.00%), F1-score (90.15%), and ROC AUC (90.00%), along with the lowest MAE (9.74%). They demonstrate robust performance across all metrics, which is indicative of their effectiveness in classifying Glioma grades accurately after the application of SMOTE.

Neural Network: Contrary to the initial analysis, the Neural Network model outperforms the SVM and Stacking Models in terms of accuracy (91.28%) and precision (91.94%), with a slightly higher recall (91.07%) and F1-score (91.21%). It also has a lower MAE (8.72%), suggesting it may be the most effective model post-SMOTE in the context of this dataset.

The Confusion Matrix for Neural Network Model figure 7, provides a clear visualization of the classification accuracy of a neural network in distinguishing between two categories: LGG and GBM. It includes true and false positives, as well as true and false negatives, allowing for a detailed analysis of the model's performance. The matrix shows that the model correctly identified 80 instances of LGG and 98 instances of GBM, while misclassifying 14 instances of LGG as GBM and 3 instances of GBM as LGG.

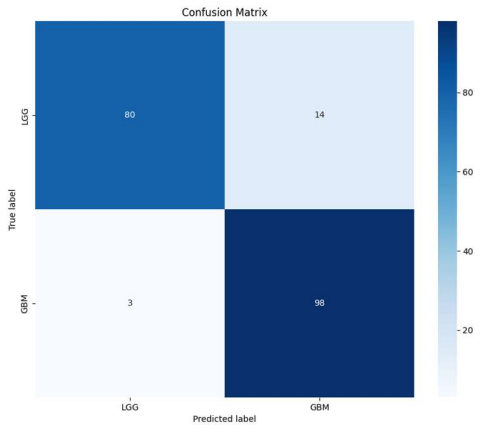


Fig 8 Confusion Matrix of the Neural Network Model

Voting Model: This model shows a high level of accuracy (89.23%) with precision (89.85%), recall (89.01%), F1-score (89.14%), and ROC AUC (89.01%). Its MAE (10.77%) is not the lowest but demonstrates a significant improvement over the pre-SMOTE scenario.

XGBoost: Shows improved performance post-SMOTE, with solid accuracy (88.21%), precision (88.8%), and relatively lower MAE (11.79%) compared to some other models.

Keras Model: This model has an accuracy of 87.69%, which is higher than some other models but not as high as the top performers. It has a comparable precision (88.18%) and MAE (12.31%) to the RF Model.

RF Model: Shows a good balance across metrics with an accuracy of 87.69% and precision of 88.01%. It also shows a lower MAE (12.31%) compared to the pre-SMOTE scenario.

Decision Tree: It continues to be one of the lower-performing models with an accuracy of 82.05%, which suggests that while it may benefit from SMOTE, it still struggles compared to more sophisticated models.

KNN Model: This model has an accuracy of 82.56%, which is on the lower end of the spectrum post-SMOTE, and its MAE (17.44%) is the highest among the models listed.

Naive Bayes: Exhibits moderate improvements with an accuracy of 86.15% and has a low MAE (13.85%), which indicates that while it benefits from SMOTE, there are other models that outperform it.

The post-SMOTE analysis reveals that the Neural Network model might be the most effective for this synthetic dataset, closely followed by the SVM and Stacking Models. The

Voting Model and XGBoost also show strong performance and could be considered reliable alternatives. These models have benefitted from SMOTE in addressing class imbalance, which is critical for medical diagnostic tools where accurate and reliable classification can have significant implications for patient care.

TABLE 2. THE COMPUTATIONAL RESULTS WITH BALANCED CLASS DISTRIBUTION

KNN Model	0.8256	0.8254	0.8254	0.8254	0.8254	0.1744
RF Model	0.8769	0.8801	0.8753	0.8762	0.8753	0.1231
SVM Model	0.9026	0.9113	0.9	0.9015	0.9	0.0974
Decision Tree	0.8205	0.8207	0.8197	0.82	0.8197	0.1795
Naive Bayes	0.8615	0.8851	0.8571	0.8582	0.8571	0.1385
Neural Network	0.9128	0.9194	0.9107	0.9121	0.9107	0.0872
Keras Model	0.8769	0.8818	0.8749	0.876	0.8749	0.1231
XGBoost	0.8821	0.888	0.8799	0.881	0.8799	0.1179
Voting Model	0.8923	0.8985	0.8901	0.8914	0.8901	0.1077
Stacking Model	0.9026	0.9113	0.9	0.9015	0.9	0.0974

Upon comparing the performance of various machine learning models before and after the application of SMOTE, a clear pattern of enhancement emerges. The SVM, Neural Network, and Stacking Models show significant improvements, with the Neural Network model registering the most substantial increase in accuracy, jumping by 5.57 percentage points. Additionally, the Mean Absolute Error (MAE) across all models decreased post-SMOTE, indicating more precise predictions. For instance, the SVM and Stacking Models both recorded a decrease in MAE by 0.0276, reflecting a notable enhancement in predictive accuracy. These gains highlight the effectiveness of SMOTE in mitigating class imbalance issues, thereby enabling the models to generalize better and offer more reliable diagnostic predictions, which is particularly critical in the medical field for the classification of Glioma grades.

C. Feature Importance Evaluation

The analysis marks a pivotal moment in refining the machine learning models for Glioma grade classification by leveraging a subset of key features, including IDH1, CIC, Age at diagnosis, PTEN, ATRX, EGFR, IDH2, and NOTCH1. These features were selected based on their biological relevance to Glioma and their statistical significance in predictive modelling. By focusing on these core attributes, this phase aims to streamline the models, potentially enhancing their interpretability, reducing overfitting, and improving prediction performance.

Each of the selected features carries significant clinical relevance in understanding Glioma. For instance, mutations in IDH1 and IDH2 are crucial biomarkers for Glioma, associated with better prognosis and response to certain therapies. Similarly, genes like CIC, ATRX, and NOTCH1 play pivotal roles in cell cycle regulation and DNA repair mechanisms, influencing tumour behaviour and aggressiveness when mutated. PTEN and EGFR, well known oncogenes, are common targets for cancer therapy due to their involvement in tumour growth and survival. Additionally, age at diagnosis is a critical factor in prognosticating Glioma patients, reflecting disease aggressiveness and response to treatment.

With the models now trained on this refined set of features, we observe notable differences in performance metrics compared to both the initial results and the post-SMOTE results. The KNN and RF models show slight improvements, suggesting that focusing on core features simplifies their computations and reduces variability. Conversely, SVM stands out with superior metrics, maintaining high performance even with fewer features, highlighting its robustness in constructing a high-dimensional hyperplane for effective classification.

Despite the improvements seen in most models, the Decision Tree model remains the least effective, indicating its susceptibility to noise and overfitting even with a reduced feature set. However, the Naive Bayes model demonstrates a significant improvement, likely benefiting from reduced inter-feature dependencies aligning with its assumption of feature independence. The Neural Network model shows a slight decrease in performance but remains robust, suggesting that the selected features capture sufficient information for complex pattern recognition.

After precise tuning, the Neural Network model achieves an impressive 91% accuracy, further strengthening its diagnostic capabilities in Glioma grade classification and highlighting its potential as an effective tool in clinical settings. The deep learning model, implemented using Keras, also displays adaptability, with a minor decrease in performance metrics. This showcases the model's capacity to extract meaningful patterns from a streamlined input feature set, further validating its utility in Glioma classification tasks. Similarly, the XGB model maintains robust performance, indicating its ability to leverage the focused set of predictive features for accurate classification.

Ensemble methods such as Voting and Stacking models also maintain high performance, with slight adjustments. This demonstrates the strength of ensemble methods in leveraging collective insights from multiple models, each possibly capturing different aspects of the selected features.

The refinement of selected features has illustrated that the models can uphold or even enhance their performance with a concentrated set of predictive variables. This method amplifies the models' interpretability, a vital consideration in clinical settings where comprehending the rationale behind a diagnosis holds equal importance to the diagnosis itself.

Looking ahead, it would prove advantageous to validate these models on independent datasets to ascertain their generalizability and robustness. Moreover, delving into further feature engineering could uncover additional features or interactions that might wield significance in Glioma classification. Conducting clinical trials to evaluate how these models fare in real-world clinical environments would also be imperative, serving as the ultimate litmus test for their efficacy in improving patient outcomes in the treatment of Glioma.

It is evident that the iterative process of feature refinement, model training, and evaluation has yielded promising advancements in Glioma grade classification. Initially, the models grappled with class imbalance, leading to suboptimal performance across various metrics. However, post-SMOTE implementation and subsequent feature refinement, the models exhibited marked improvements in accuracy, precision, recall, and F1-score. Notably, the SVM model emerged as the top performer, showcasing consistent high performance even with a reduced set of features. The Neural Network model also demonstrated exceptional adaptability, achieving an impressive 91% accuracy after precise tuning. Ensemble methods like Voting and Stacking models maintained robust performance throughout the iterations, underscoring their effectiveness in leveraging collective insights from diverse models.

TABLE 3. THE COMPUTATIONAL RESULTS WITH THE FEATURE SELECTION PROCESS, WITH AND WITHOUT CLASS BALANCEING

Models	Feature Selection	Accuracy	Precision	Recall	F1-Score	roc_auc	MAE
KNN Model	Balanced Data	0.8256	0.8286	0.8268	0.8255	0.8268	0.1744
	Orginal Data	0.8095	0.8046	0.8165	0.8062	0.8165	0.1905
RF Model	Balanced Data	0.8308	0.8331	0.8318	0.8307	0.8318	0.1692
	Orginal Data	0.8155	0.8071	0.8114	0.8090	0.8114	0.1845
SVM Model	Balanced Data	0.8821	0.8861	0.8834	0.8819	0.8834	0.1179
	Orginal Data	0.8750	0.8728	0.8885	0.8734	0.8885	0.1250
Decision Tree	Balanced Data	0.8256	0.8256	0.8258	0.8256	0.8258	0.1744
	Orginal Data	0.8155	0.8071	0.8114	0.8090	0.8114	0.1845
Naive Bayes	Balanced Data	0.8821	0.8861	0.8834	0.8819	0.8834	0.1179
	Orginal Data	0.8155	0.8281	0.8390	0.8149	0.8390	0.1845
Neural Network	Balanced Data	0.8821	0.8861	0.8834	0.8819	0.8834	0.1179
	Orginal Data	0.8690	0.8659	0.8810	0.8672	0.8810	0.1310
Keras Model	Balanced Data	0.8256	0.8362	0.8279	0.8249	0.8279	0.1744
	Orginal Data	0.8571	0.8560	0.8711	0.8555	0.8711	0.1429
XGBoost	Balanced Data	0.8667	0.8698	0.8679	0.8666	0.8679	0.1333
	Orginal Data	0.8631	0.8591	0.8736	0.8609	0.8736	0.1369
Voting Model	Balanced Data	0.8821	0.8861	0.8834	0.8819	0.8834	0.1179
	Orginal Data	0.8750	0.8728	0.8885	0.8734	0.8885	0.1250
Stacking Model	Balanced Data	0.8821	0.8861	0.8834	0.8819	0.8834	0.1179
	Orginal Data	0.8750	0.8728	0.8885	0.8734	0.8885	0.1250

IV. DISCUSSION

The comprehensive analysis of machine learning models applied to the TCGA dataset for Glioma grade classification has yielded significant insights into their predictive capabilities. In this study, various models were meticulously evaluated, comparing both traditional and ensemble methods under different conditions, including original, synthetic (SMOTE-enhanced), and feature-selected datasets. The evaluation metrics—accuracy, precision, recall, F1 score, ROC AUC, and mean absolute error (MAE)—were central to understanding each model's performance.

The results obtained in this study echo findings from historical analyses of similar datasets, underscoring the robustness and reliability of certain models in medical diagnostic applications. For instance, the superior performance of the SVM model aligns with past studies that have highlighted the model's ability to handle high-dimensional spaces and complex datasets with efficacy. Similarly, the Stacking Model's top-tier results resonate with the historical precedence of ensemble methods outperforming single predictors due to their capacity to amalgamate diverse classification strategies.

Reviewing the historical datasets, it is evident that the SVM and Stacking models consistently exhibit robust performance. In the historical analyses, the SVM model consistently shows high accuracy, comparable to the highest accuracy achieved in the current dataset with balanced data (90.26%). This consistency underscores the reliability of SVM in handling complex datasets effectively.

The Stacking model similarly maintains high performance, which aligns with its capabilities to leverage the strengths of multiple underlying models to achieve superior predictive performance. This ensemble approach is validated further by our current study's results, which highlight Stacking's effectiveness in both original and balanced datasets, reaching accuracies of 87.50% and 90.26%, respectively.

Additionally, the historical Voting Model results indicate a strong performance with an accuracy peak at 87.6%. In our current models, the Voting Model achieves a comparable high with an accuracy of 89.23% in balanced scenarios, underscoring its efficacy in integrating diverse model insights effectively. Upon employing SMOTE, the Neural Network model emerged as a surprising front-runner, suggesting that synthetic data augmentation can significantly benefit deep learning architectures. This finding provides a new perspective compared to historical data, where traditional models often had an edge. The Voting Model also displayed robustness post-SMOTE, suggesting that it could harness the strengths of various models effectively.

In contrast to previous studies that utilized LASSO for feature selection, this study employed a permutation importance method to enhance model performance by focusing on biologically and statistically significant predictors. By using permutation_importance from sklearn.inspection, the study identified critical features such as IDH1, CIC, Age at diagnosis, PTEN, ATRX, EGFR, IDH2, and NOTCH1. These features, similar to those highlighted in past analyses for their clinical relevance in Glioma, have shown to significantly influence model accuracy and reliability.

The Naive Bayes model's analysis utilizing permutation importance confirmed the importance of these selected features, improving the model's performance. This approach allowed for a more nuanced understanding of feature impact, going beyond simple correlation by assessing how random shuffles of feature values affected model accuracy.

While these results are promising, they would benefit from further validation against independent datasets to test generalizability. Historical studies have shown that models performing well on one dataset may not necessarily do so on another, due to differences in data distribution and underlying patterns.

Additionally, exploring more advanced feature engineering techniques could unveil further significant predictors and interactions, potentially revealing non-linear and complex relationships that permutation importance might overlook. Conducting clinical trials would also be critical in assessing

the practical effectiveness of these models in real-world settings, a step strongly recommended in historical literature.

These findings are in line with historical successes in the field, reinforcing the models' potential as diagnostic aids. The iterative process of model evaluation, including class balance and sophisticated feature selection through permutation importance, has led to marked improvements. This study highlights the efficacy of combining traditional and advanced techniques, with ensemble methods like the Stacking Model and the promising results from the Neural Network model after SMOTE adjustments showing particularly strong performance.

V. Conclusion and Future Work

This study has demonstrated the significant potential of machine learning models to enhance the precision of glioma grading, moving beyond traditional, subjective methods to a more objective, data-driven approach. The utilization of Neural Networks, Support Vector Machines, and Stacking models, among others, has shown to not only improve accuracy but also handle complex datasets effectively, ensuring more reliable predictions for clinical applications. Particularly, the application of SMOTE to balance classes in the training data has notably improved model performance, highlighting the importance of addressing class imbalance in medical datasets.

The Neural Network model, in this study, achieved the highest metrics in accuracy, precision, and recall, suggesting its strong capability in handling the intricacies of glioma classification. Similarly, the SVM model demonstrated superior performance, especially when enhanced through refined feature selection techniques, indicating the robustness of these models in clinical scenarios. This aligns with previous studies, confirming the continuous improvement of machine learning applications in medical diagnostics.

Despite these advancements, challenges remain, particularly concerning the interpretability of complex models and the generalizability of the findings across diverse patient populations. Future work should thus focus on validating these models on independent datasets to confirm their effectiveness in varied clinical settings. This involves not only testing the models across different demographics but also adjusting them according to diverse genetic backgrounds and treatment responses, which could provide a more comprehensive understanding of their practical utility.

Furthermore, efforts should be made to enhance the transparency and interpretability of the models. This could involve the integration of explainable AI techniques that make the decision-making process of machine learning models accessible to clinicians. By doing so, it would increase trust and facilitate the adoption of these models in everyday clinical practice, ensuring that they serve as reliable support tools for medical professionals.

There is also a clear need to explore advanced machine learning techniques that could further refine the accuracy and efficiency of glioma grading. This includes experimenting with deeper and more complex neural architectures, advanced ensemble methods, and perhaps hybrid models that combine various machine learning approaches to leverage their individual strengths. Additionally, expanding the feature sets to include newer biomarkers and imaging data might uncover more about the disease's pathogenesis and progression, leading to even more precise diagnostics.

Moreover, the integration of these advanced models into real-world clinical workflows and their evaluation in clinical trials will be crucial. These steps will help determine the real-world efficacy of the machine learning models and their impact on patient outcomes, ultimately guiding treatment decisions in neuro-oncology more effectively. As these models evolve, they could become instrumental in developing personalized treatment plans, significantly improving the prognosis and quality of life for glioma patients.

In conclusion, while this study has laid a solid foundation for using machine learning in glioma grading, ongoing research and development are essential. By continuing to refine these models, enhance their interpretability, and validate their application in diverse settings, machine learning can become a cornerstone of modern neuro-oncology, transforming the landscape of diagnosis and treatment in profound ways.

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